CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of M. Pharm (Quality Assurance)

University Theory Examination January 2018

PHQA001 MODERN ANALYTICAL TECHNIQUES

Date: 01.01.2018 , Monday Time: 10.00 a.m. to 01.00 p.m Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

Q.1	**A)**	Explain the basic elution principle for column and planar chromatography.	**05**
	B)	Give classification of stationary phase used in Gas chromatography (GC) and explain the factors governing the selection of stationary phase in GC.	**05**
Q.2	**A)**	Write a note on factors affecting separation in HPLC.	**05**
	B)	Describe in brief mobile phase selection and optimization for HPTLC.	**05**
		OR	
Q.2	**A)**	Explain the fragmentation rules in mass spectroscopy.	**05**
	B)	Discuss the principle of affinity and size exclusion chromatography.	**05**
Q.3		**Answer the following (Any three)**	**15**
	(A)	Differentiate between DSC and DTA.	
	(B)	What is MALDI? Explain its working in brief.	
	(C)	Differentiate between GC and HPLC.	
	(D)	Wrtie a note on capillary electrophoresis.	

SECTION-II

Q.4	**(A)**	Discuss different types of electronic transitions observed in spectroscopy.	**05**
	(B)	Explain different factors affecting absorption maxima and absorption spectra in UV-visible spectroscopy.	**05**
Q.5	**(A)**	Write note on Derivative spectroscopic method used for analysis of combination drugs.	**05**
	(B)	Discuss Michaleson Interferometer and sampling techniques used in IR spectroscopy.	**05**
		OR	
Q.5		What is Fluorescence spectroscopy? Discuss the instrumentation of fluorescence spectroscopy. Write factors affecting fluorescence spectra.	**10**
Q.6		**Answer the following (Any three)**	**15**
	(A)	Write short note on: 2D NMR technique.	
	(B)	What is X-ray diffraction technique? Discuss Bragg's law.	
	(C)	Write note on ELISA.	
	(D)	Write a note on ^{13}C NMR technique.	

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